Transverse vs. axial driving in the radical-pair mechanism

Where the low-field / high-field regimes divide, and what sets the optimum on the transverse side

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Summary

1. **The crossover is real and sharp.** Transverse-hyperfine driving gives way to axial driving at $B_0 \approx 0.6$ mT $\approx A_{zz}/3$ — set by the axial coupling, not the transverse one.

2. **The transverse coupling has an interior optimum.** More transverse coupling does not monotonically help; both spike amplitude and angular sensitivity turn over, and the optimum slides with coherence time.

3. **That optimum obeys a clean scaling law,** $b_{opt} = \frac{C \cdot A_{zz}}{\sqrt{\tau}}$. Inverting it, the implied optimal lifetime depends *only* on the dimensionless hyperfine shape ratio $x = \frac{a_{iso}}{T}$ — a generic property of aromatic nitrogen. Real FAD N5 implies $\tau^* \approx 27$ μs.

1. Where does transverse driving hand over to axial driving?

We have been working in the high-field regime, where B_0 drives singlet–triplet dynamics through axial Zeeman coupling to the electrons. The quantum-needle work shows that at geomagnetic fields the mechanism is different: the field acts *through* the transverse anisotropic hyperfine components of strongly-coupled nitrogens, producing a sharp orientational feature at long coherence times. The obvious question is where the handover happens, and whether it is gradual or abrupt.

Experimental Setup

- Toy RPM: [X• Y•] pair, τ = 10 μs. X• carries an N5-like ¹⁴N tensor (−0.0989, −0.0989, 1.7569) mT; Y• is a purely axial Trp-N1-like ¹⁴N (0, 0, 1.0812) mT. All transverse coupling therefore lives on X•.
- B_0 swept 0.05 → 100 mT, rotated through θ. Exact singlet yield Φ_S; angular sensitivity defined as

$$\Gamma = max_\theta \left| \frac{\partial \Phi_S}{\partial \theta} \right|$$

- Diagnostic: the ablation fraction

$$f = 1 - \frac{\Gamma_{transverse\ zeroed}}{\Gamma_{full}}$$

- f ≈ 1 means transverse driving; f ≈ 0 means the transverse components are irrelevant.
- No exchange or dipolar e–e coupling, no relaxation.

Result

Two independent signatures flip together between 0.5 and 0.7 mT. Below the crossover, zeroing the transverse components destroys ~95% of the angular sensitivity and the sharp feature is locked at θ = 90°. Above it, removing them changes nothing — and at 100 mT the sensor is *better* without them (f = −1.23), i.e. transverse coupling has become a blurring term. The feature simultaneously unpins from 90°, jumping to 38.6° and then drifting back as B_0 rises.

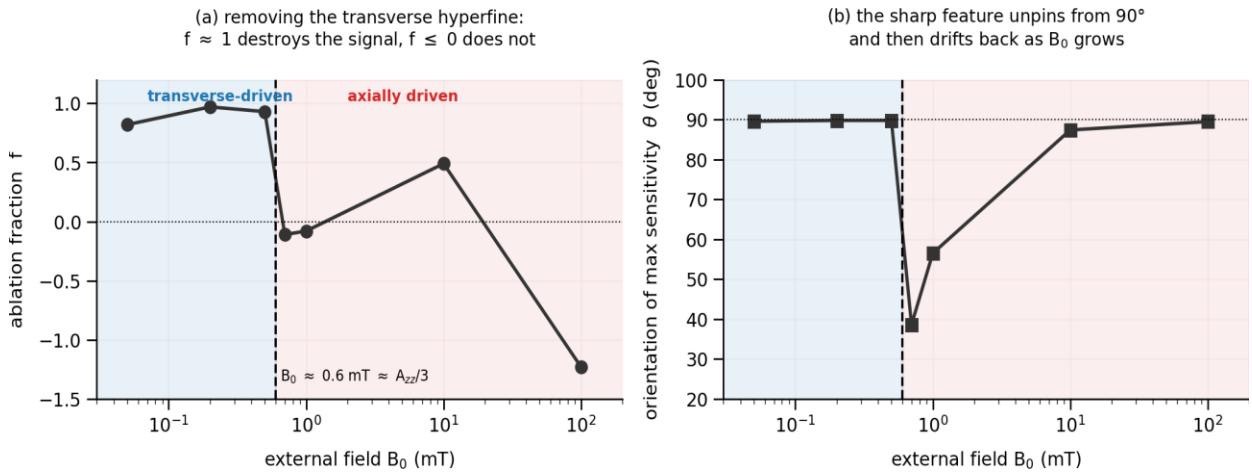


Figure 1 — The regime change at B_0 ≈ 0.6 mT ≈ Azz/3. (a) Transverse ablation stops mattering. (b) The sharp feature unpins from 90°.

The crossover scale is set by the dominant **axial** coupling ($A_{zz}/3$), not the transverse one. Physically: below it, the Zeeman term perturbs an avoided crossing of gap 2b and the needle sits at exactly 90°; above it, S–T± are Zeeman-split away and the surviving mixing comes from the orientation-dependent axial hyperfine projection along B_0, which needs no transverse coupling at all.

Caveats. The crossover is only bracketed on 0.5–0.7 mT, not finely resolved, and its lifetime dependence has not been tested. The high-field resonances are <0.1° wide and would be washed out by realistic relaxation or librational motion — this is a statement about Hamiltonian structure, not an observable compass response. Two engines agree to 1e−14 at high field.

2. On the transverse side, is more coupling always better?

Result 1 implies it cannot be. The quantum-needle work reports the spike growing with transverse coupling, but that growth has to stop — at the very least it must fail once B_0 is no longer the weak-field perturbation. So we asked what the optimum actually looks like.

Experimental Setup

- Same two-nucleus system. λ scales the transverse components only, A_{zz} held fixed; the control variable is the physical transverse coupling

$$b = \gamma_e \sqrt{A_{xx}^{\,2} + A_{yy}^{\,2}}$$

- λ = 1 → b = 140 μT; real N5 sits at λ ≈ 0.95, b = 133 μT — already ~2.7× Earth's field.
- B_0 = 50 μT fixed. 17 λ values × τ ∈ {1, 10, 100} μs. Exact Φ_S, no approximation.
- Gates passed: $\Phi_S(H = 0) = 1.000000$; spike vanishes identically at zero transverse coupling (confirming a true level crossing); spike deepens with τ.

Result

There is an **interior optimum** — both amplitude and sensitivity rise then fall. The earlier monotone impression came from sweeping only to ×4 from a smaller base, which hid the turnover. Critically, the optimum is neither at Earth's field nor at the recombination rate: it sits at several × B_{earth} and it **slides with τ**. It is a transverse coupling × coherence-time trade-off, not a resonance with the geomagnetic field.

τ	amplitude optimum	sensitivity optimum	spike present?

1 μs	—	—	no
10 μs	378 μT	224 μT	yes
100 μs	175 μT	70 μT*	yes

*resolution-limited at τ = 100 μs (feature FWHM < 0.7° against a 0.05° grid); the amplitude optima are robust.

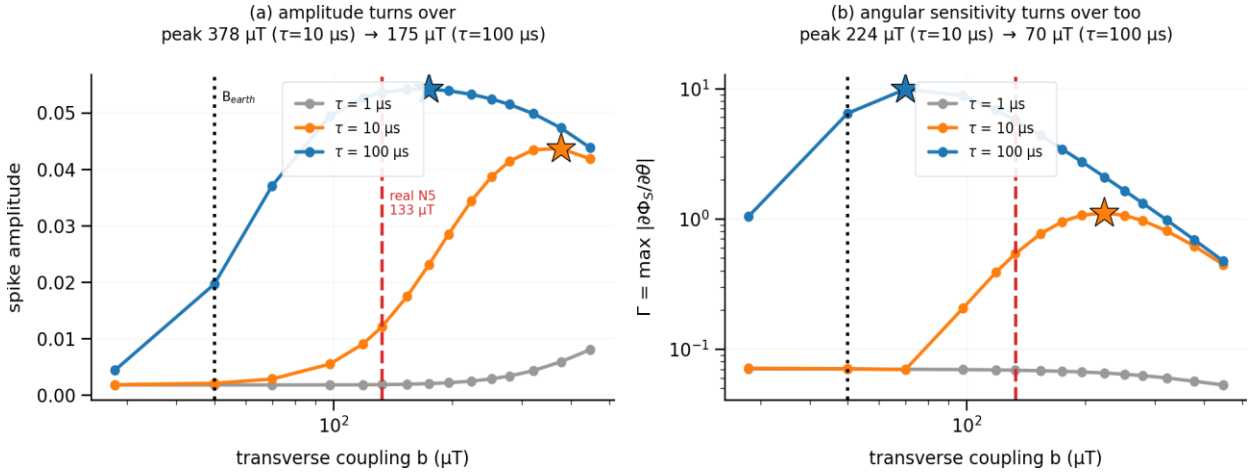


Figure 2 — The transverse coupling has an interior optimum, and it moves with coherence time. Stars mark the optima; dashed red line is real FAD N5.

This retires the 'hyperfines tuned to Earth's field' reading: the couplings that set the spike are already ~2× Earth, and the sensitivity optimum wants them *higher*, not lower. What survives is narrower and more interesting — real N5 at 133 μT stays within a factor ~2 of the sensitivity optimum across the whole physiological τ range.

3. Modelling the optimum — and where it points

The optimum moving systematically with τ suggested a law rather than a coincidence, so we mapped the optimum transverse coupling for angular sensitivity, $b_{opt}(\tau)$, across three axial couplings to further explore the relationship between axial and transverse couplings.

Experimental Setup

- For each axial hyperfine and coherence time (A_{zz}, τ) cell: sweep b, compute $\Gamma = \max |\partial\Phi_S/\partial\theta|$ at each b, take the argmax. 3 × 5 = 15 cells.
- $A_{zz} \in \{0.6046, 1.2, 1.7569\}$ mT — real FAD N10, a synthetic midpoint, and real FAD N5. $\tau \in \{5, 10, 20, 50, 100\}$ μs.
- Tracks the sensitivity optimum (not the amplitude one); only this branch obeys a clean law.

Result — the scaling law

$$b_{opt} = \frac{C \cdot A_{zz}}{\sqrt{\tau}}$$

with C ≈ 391 μT·√μs per mT of A_{zz}

Normalising by A_{zz} collapses a 3× range of axial coupling onto a single curve, agreeing to four decimal places (raw b_{opt} at τ = 10 μs differs by 3×: 75.2 / 149.1 / 218.3 μT). Fitted exponents are −0.4952, −0.4954, −0.4955 — so −1/2 holds to ~1%, with a ~2% residual drift in C across the τ range. **Azz sets the scale and nothing else does.**

Inverting the law for b = 132.5 μT gives τ ≈ 29 μs. Running the Γ sweep at that lifetime, real FAD N5 lands at **100.0% of peak**.

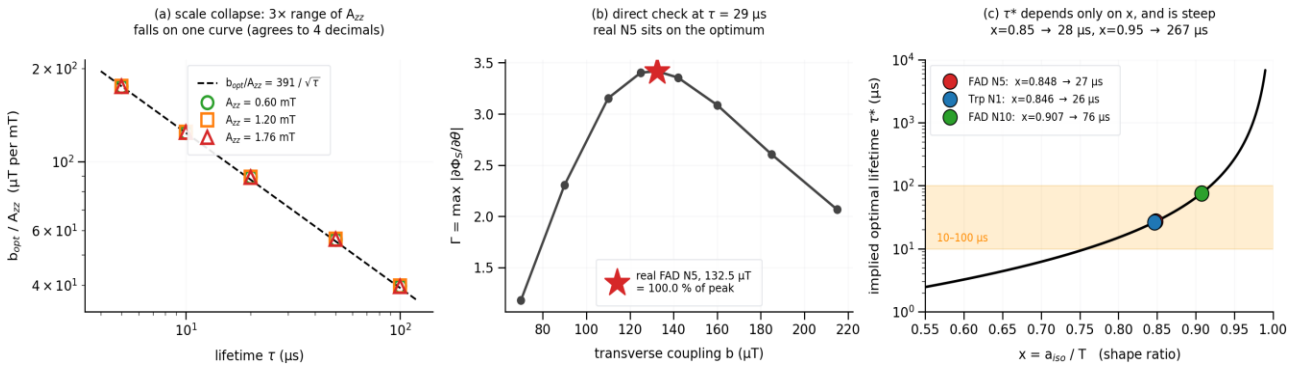


Figure 3 — (a) the scale collapse; (b) the direct check at τ = 29 μs; (c) the implied optimal lifetime as a function of hyperfine shape ratio.

So we've now developed a scaling law that describes the motion of the optimal transverse coupling with coherence times. We've then shown that this scaling is determined purely by the axial hyperfine coupling, A_{zz}. We now move on to link the chemical shape of aromatic nitrogens to this axial–transverse coupling relationship to show that this chemical shape factor alone can describe the optimal angular sensitivity of singlet yield of the radical pair mechanism.

Why this connects to aromatic nitrogen

Writing the tensor in axial form,

$$A_{zz} = a_{iso} + 2T\ , \quad b = \sqrt{2}\,|\,a_{iso} - T\,|$$

Setting the real b equal to the optimal b and solving for the lifetime, T cancels, leaving

$$\tau^*(x) = \left[\frac{k\,(x+2)}{\sqrt{2}\,|\,x-1\,|}\right]^2$$

with x = a_{iso}/T and k = C/1000

The cancellation is only legal because of the collapse — otherwise the coupling magnitude would survive into τ* and this would be a statement about one nucleus rather than about tensor *shape*.

nucleus	aiso (mT)	T (mT)	x = aiso/T	implied τ*
FAD N5	0.5233	0.6168	0.848	27 μs
Trp N1	0.3215	0.3798	0.847	26 μs
FAD N10	0.1887	0.2080	0.907	76 μs

All three are π-conjugated sp² aromatic nitrogens and all three share x ≈ 0.85–0.91 — the unpaired spin density sits in a p_z orbital, which fixes the dipolar term relative to the contact term. So the tens-of-μs figure is not a tuned parameter; it is what aromatic nitrogen chemistry hands you. The needle's optimal lifetime is set by hyperfine **shape**, not by magnitude.

Caveats. τ*(x) is steep near x → 1 (x = 0.85 → 28 μs, x = 0.95 → 267 μs), so this predicts the band, not the number. A 300,000-sample null over plausible (a_{iso}, T) ensembles shows landing in 10–100 μs is *not* rare — the agreement is mechanistic, not evidence of evolutionary optimisation. The collapse was tested over only a 3× range in A_{zz}.

Where this goes next

- Lifetime dependence of the crossover — does B_0^* move with τ, and does either regime survive physical relaxation rates? This links the two studies directly: E7 says the weak-field optimum is a $b \cdot \sqrt{\tau}$ budget, E5 says the boundary is set by A_{zz}. Resolve the 0.5–0.7 mT bracket finely at the same time.
- Full bath, real metric. Everything above uses a two-nucleus toy with max-slope at a single orientation. The authoritative test is the detector-noise / Fisher-information precision over the full $\Phi_S(\theta,\varphi)$ sphere with rotational averaging, on the 14–16-spin system. Decision criterion: does the precision optimum keep tracking τ (the coherence-budget story), or pin near 50–130 μT independently of τ (a genuine tuning result)?
- Add electron–electron coupling. Exchange and dipolar terms are zero in all three experiments, and a related window effect drops to 0.2% of its coupling-free value at r = 2 nm. This is the largest single gap.